Peak water has passed

Summer runoff from the Sorne glacier catchment to 2060

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−38%

August runoff in the 2050s, compared with 1991–2020

Median of 600 simulations, SSP2-4.5. 10–90% range: −49% to −27%.

Why this catchment

- Sorne catchment: 74 km², 31% glacierised in 2020
- In August, melt from the glacier supplies 62% of the river's flow
- Feeds the Lac de Brenay reservoir (hydropower, head 480 m)
- Irrigation for 1,900 ha of orchards and vineyards downstream
- The last runoff projection for the Sorne is from 2015 and used CMIP5

Study area and data

Discharge	Gauge at Pont-de-Sorne (1,480 m a.s.l.), daily, 1981–2024
Weather	3 AWS at 1,620 m, 2,340 m and 2,910 m: T and P, daily
Mass balance	14 stakes on the Glacier de Sorne, 2009–2024
Glacier outlines	1985, 2003, 2016, 2023 (orthophotos)
Ice thickness	GPR survey, 2018

Methods (1): melt

$$M = DDF \cdot max(T - T_0, 0)$$

- M: melt in mm w.e. d^{-1}; T: daily mean air temperature of the elevation band
- DDF_{snow} = 4.1 ± 0.6 mm w.e. $°C^{-1}$ d^{-1}
- DDF_{ice} = 7.3 ± 0.9 mm w.e. $°C^{-1}$ d^{-1}
- T_0 = 0 °C; lapse rate −5.8 °C km^{-1}; 100 m elevation bands
- Rain or snow: single threshold at 1.0 °C

Degree-day approach after Hock (2003). ± is one standard deviation across the 50 retained parameter sets.

Methods (2): glacier and routing

- Glacier geometry updated once a year with the Δh-parameterisation (Huss et al., 2010)
- Melt and rain routed through two linear reservoirs:

$$Q(t) = S_{fast}(t) / k_{fast} + S_{slow}(t) / k_{slow}$$

- k_{fast} = 2.1 d, k_{slow} = 38 d
- Evapotranspiration: Hamon, off-glacier area only
- 9 free parameters in all

Calibration and validation

Daily discharge	Calibration 1981–2005	Validation 2006–2024
NSE	0.87	0.82
log-NSE	0.84	0.79
Volume bias	+1.2%	−3.4%

Glacier mass balance, 2009–2024: RMSE 0.31 m w.e. a^{-1} against the stakes.

Monte Carlo: 20,000 parameter sets sampled; the 50 best on a combined discharge and mass-balance score were kept.

Climate forcing

- 12 GCMs from CMIP6
- 3 scenarios: SSP1-2.6, SSP2-4.5, SSP5-8.5
- Bias-corrected to the three stations by quantile mapping, reference 1991–2020
- 12 GCMs × 50 parameter sets = 600 simulations per scenario, 1,800 in all

Peak water

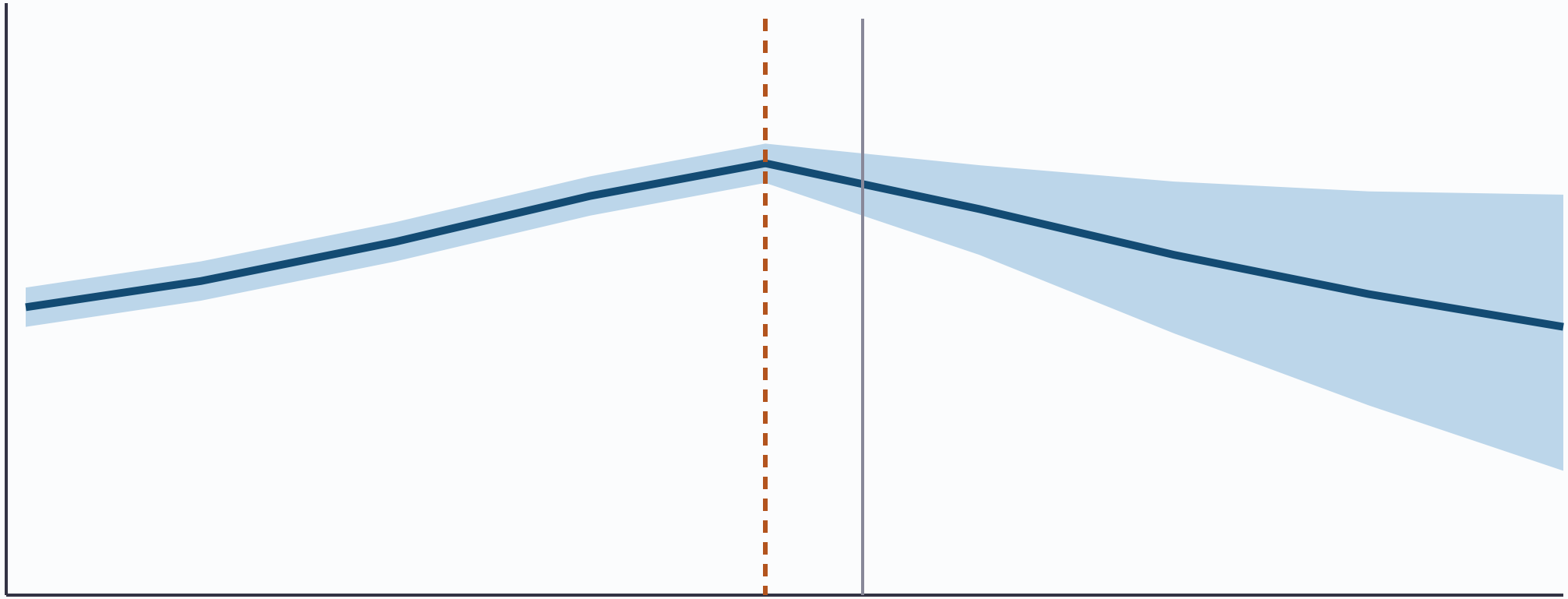


Figure 2. Annual runoff at Pont-de-Sorne, 1981–2060, SSP2-4.5. Line: median. Band: 10–90% range of 600 simulations. Dashed line: peak water, around 2019 (range 2012–2027). Grey line: end of observations.

August runoff, change from 1991–2020

Scenario	2031–2040	2051–2060
SSP1-2.6	−14% [−22, −6]	−29% [−41, −18]
SSP2-4.5	−17% [−26, −8]	−38% [−49, −27]
SSP5-8.5	−19% [−30, −9]	−47% [−60, −35]

Median [10th, 90th percentile] of 600 simulations per scenario.

The year as a whole changes much less

- Annual runoff: −9% by the 2050s (SSP2-4.5, median)
- April–June runoff: +21%
- Snow melts about three weeks earlier, and more winter precipitation falls as rain
- The month of highest flow moves from July to June

The water does not disappear. It arrives earlier.

The glacier itself

	2020	2060 (SSP2-4.5, median)
Area	22.9 km²	11.6 km²
Ice volume	0.71 km³	0.24 km³

Two thirds of the ice volume is gone by 2060. Under SSP5-8.5: 0.15 km³ left.

Where the spread comes from



Figure 4. Share of the variance in August runoff, 2051–2060 (ANOVA). GCM: 46%. Scenario: 31%. Hydrological model parameters: 17%. Internal variability: 6%.

What it means for Lac de Brenay

- July–September inflow: 41 hm³ today (1991–2020 mean)
- 2051–2060: 29 hm³ (SSP2-4.5, median)
- Spring inflow rises, but the reservoir is usually full by mid-June and spills

1 hm³ = 1 million m³. Inflow computed at the dam, 9 km downstream of the gauge.

Sensitivity

What was varied	August 2051–2060, SSP2-4.5
DDF_{ice} ± 1 standard deviation	−34% to −42%
Lapse rate −5.3 to −6.3 °C km^{-1}	−36% to −40%
Rain/snow threshold 0 to 2 °C	−37% to −39%

Limitations

- No debris cover: 12% of the ablation area is debris-covered today, and debris slows melt
- Degree-day factors are held constant to 2060
- No contribution from permafrost or rock glaciers
- Quantile mapping assumes model biases stay the same in a warmer climate
- One catchment: do not scale these numbers to the whole canton

Conclusions

- Peak water in the Sorne is behind us (around 2019)
- August runoff falls by about a third by the 2050s under a middle scenario
- The annual total barely moves: the question is storage and timing, not scarcity

Code, data, funding

- Model code: git.valmont.example/hydro/sorne-runoff (MIT licence)
- Station and stake data: on request from the Canton's water office
- Preprint under review at the Journal of Alpine Hydrology
- Funded by the Fonds Valmont pour la recherche alpine, grant 21-0457

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Thank you

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